Oligopoly

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 18, Oligopoly.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 18, "Oligopoly".

What you must know from this chapter

1. **Conditions:** what makes a market an oligopoly.
2. **Cartels and collusion:** firms acting together as one monopolist.
3. **Why collusion breaks down:** each firm's best response, and the Nash equilibrium.
4. How the outcome under oligopoly compares with monopoly and perfect competition.

Part 1

What Makes a Market an Oligopoly?

Oligopoly

Definition: Oligopoly
A market with only a few sellers, each offering a product similar or identical to the others'.

- **Few sellers:** a handful of firms supply most of the market.
- **Similar or identical products.**
- **Barriers to entry:** new firms find it hard to enter.
- **Interdependence:** each firm's profit depends on what its rivals do.

Examples: commercial aircraft (Boeing and Airbus); U.S. wireless phone service.

Definition: Duopoly
An oligopoly with exactly two sellers. This lecture studies a duopoly.

Real-world examples

- **Commercial aircraft:** Boeing and Airbus build almost all large passenger jets.
- **U.S. wireless phone service:** a few national carriers serve most customers.
- **AI model builders** (OpenAI, Anthropic, Google, Meta): they fit all four conditions.
  - **Few sellers:** a handful of firms build the leading models.
  - **Similar products:** their assistants do much the same tasks.
  - **Barriers to entry:** the cost of training a leading model has grown 2 to 3 times a year since 2016.
  - **Interdependence:** when one firm cuts its price or releases a better model, its rivals lose customers, so each must allow for what the others do.

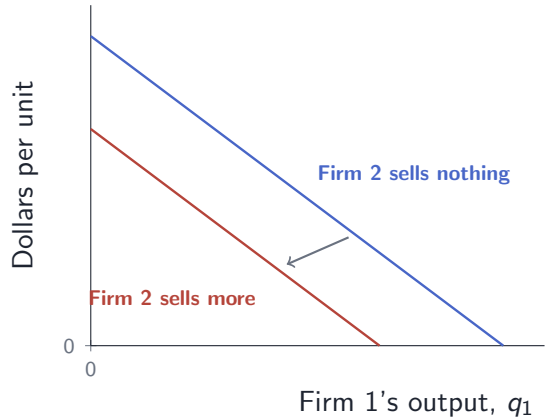
Finding

The AI applications built on these models are monopolistic competition (Chapter 17). The model builders themselves are an oligopoly.

Source: B. Cottier et al., "How Much Does It Cost to Train Frontier AI Models?", Epoch AI, 2024.

Interdependence

- The two firms sell the same product at one market price.
- The price depends on the output of *both* firms.
- So the more firm 2 sells, the less demand is left for firm 1.
- Each firm must allow for what the other does.



Four market structures

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Sellers	Many	Many	**A few**	One
Product	Identical	Differentiated	**Similar or identical**	No close substitutes
Entry	Free	Free	**Difficult**	Blocked
Rivals matter?	No	Little	**Yes**	No rivals

Finding

What sets oligopoly apart is interdependence: each firm must allow for what its rivals do.

Part 1 summary

- Oligopoly: a few sellers of similar or identical products.
- Entry is difficult.
- Interdependence: each firm's profit depends on its rival's output.
- The more one firm sells, the less demand is left for the other.
- Compared with perfect competition and monopoly: there, no rival's choice enters a firm's profit.

Check: Part 1

Which of the following examples illustrates an oligopoly market?

A A farmers' market with many individuals selling sweet corn and tomatoes

B A city whose electrical service is provided by one electric co-operative

C A city with two firms that are licensed to sell school uniforms for the local schools

D A city with many independently owned hair styling salons

Check: Part 1 – answer

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Finding

Two sellers of the same product, with entry blocked by licensing: a duopoly. A is perfect competition, B is monopoly, D is monopolistic competition.

Part 2

Collusion: Acting as One Monopolist

What collusion means

Definition: Collusion

An agreement among the firms in a market on the output each produces and the price they charge.

Definition: Cartel

A group of firms acting together. A central body sets total output for all of them.

- The cartel chooses output to maximize the firms' *joint* profit.
- So the firms act as one monopolist.

Why the cartel can do better

- When firm 1 sells more, the market price falls.
- The lower price applies to firm 2's units too. That is a cost firm 1 imposes on firm 2.
- Acting alone, firm 1 ignores that cost: it is firm 2's loss, not firm 1's.
- The cartel counts it, because it maximizes the two firms' profit together.

Finding

Collusion makes each firm take account of the cost its output imposes on its rival. The firms then behave exactly like a monopolist.

The model

Two firms, 1 and 2, sell an identical product.

- **Market demand:**
$$P = 1 - Q, \qquad Q = q_1 + q_2.$$
- **Costs are zero:** TC $= 0$, so MC $= 0$.
- So each firm's profit is its revenue: $\pi_i = P \times q_i$.

Finding

Any $P = a - bQ$ with MC $= c$ becomes $P = 1 - Q$ with MC $= 0$ by a choice of units. So the simple model shows exactly what the general one shows. We use it to make the exposition easier.

Recall from Chapter 16: MR for a straight-line demand curve

From the monopoly lecture, marginal revenue is $\text{MR} = P + Q \times \dfrac{\Delta P}{\Delta Q}$. For $P = a - bQ$:

$$
\begin{aligned}
\text{MR} &= P + Q \times \frac{\Delta P}{\Delta Q} && \text{output effect plus price effect} \\
&= (a - bQ) + Q \times \frac{\Delta P}{\Delta Q} && \text{replace } P \text{ with } a - bQ \\
&= (a - bQ) + Q \times (-b) && \text{each extra unit lowers } P \text{ by } b, \text{ so } \tfrac{\Delta P}{\Delta Q} = -b \\
&= a - bQ - bQ && Q \times (-b) = -bQ \\
&= a - 2bQ && -bQ - bQ = -2bQ
\end{aligned}
$$

Finding

MR starts at the same point as demand and is twice as steep. Here $a = 1$ and $b = 1$, so $\text{MR} = 1 - 2Q$.

The cartel's output and price

A cartel acts as a single monopolist. It does two things:

1. finds the output that maximizes joint profit, where MR = MC;
2. uses a rule to allocate that output between the two firms.

$1 - 2Q = 0$	set MR = MC = 0
$Q_K = \frac{1}{2}$	add $2Q$ to both sides; divide by 2
$P_K = 1 - \frac{1}{2} = \frac{1}{2}$	read the price from demand

Q_K is the collusive output: the monopoly output.

Notation: K stands for cartel. The letter C is kept for competition, as in earlier chapters.

Splitting the output and the profit

The two firms are identical, so they split the cartel output equally.

$$q = \tfrac{1}{2} \times Q_K = \tfrac{1}{4} \qquad \text{each firm's output}$$

$$\pi = P_K \times q = \tfrac{1}{2} \times \tfrac{1}{4} = \tfrac{1}{8} \qquad \text{each firm's profit}$$

Finding

Each member produces $\frac{1}{4}$ and earns $\frac{1}{8}$: half the monopoly profit of $\frac{1}{4}$.

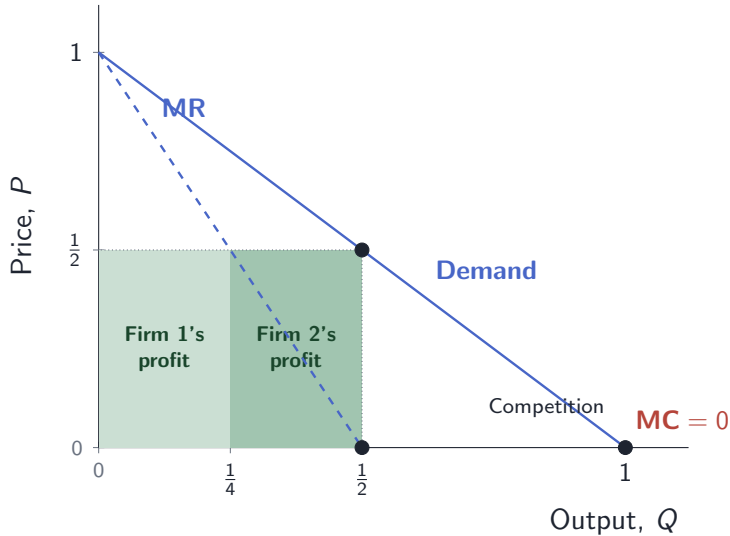
The benchmark: perfect competition

- Under perfect competition, price equals marginal cost: $P = \text{MC} = 0$.
- From demand: $1 - Q = 0$, so $Q = 1$.
- Profit is zero.

Finding

The cartel produces $\frac{1}{2}$, half the competitive output of 1, and charges $\frac{1}{2}$, above marginal cost.

The cartel on the graph



Part 2 summary

- Collusion: the firms agree on output and price; a cartel sets total output.
- The cartel counts the cost each firm's output imposes on the other, so it acts as one monopolist.
- Model: $P = 1 - Q$, $\text{MC} = 0$; any straight-line demand with constant MC reduces to it.
- $\text{MR} = 1 - 2Q = 0$ gives $Q_K = \frac{1}{2}$ and $P_K = \frac{1}{2}$. Each firm makes $\frac{1}{4}$ and earns $\frac{1}{8}$.
- Compared with perfect competition ($Q = 1$, $P = 0$): half the output, and a price above MC.

Check: Part 2

If duopoly firms that are not colluding were able to successfully collude, then

A price and quantity would rise.

B price and quantity would fall.

C price would rise and quantity would fall.

D price would fall and quantity would rise.

Check: Part 2 – answer

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A price and quantity would rise.

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Finding

A cartel restricts output to the monopoly level, so quantity falls and price rises.

Part 3

Oligopoly Without Collusion

The key insight: price depends on both outputs

- Now there is no cartel. Each firm chooses its own output to maximize its own profit.
- The price depends on total output: $P = 1 - Q = 1 - q_1 - q_2$.
- So firm 1's profit depends on its own output *and* on firm 2's:

$$\pi_1 = P \times q_1 = (1 - q_1 - q_2) \times q_1.$$

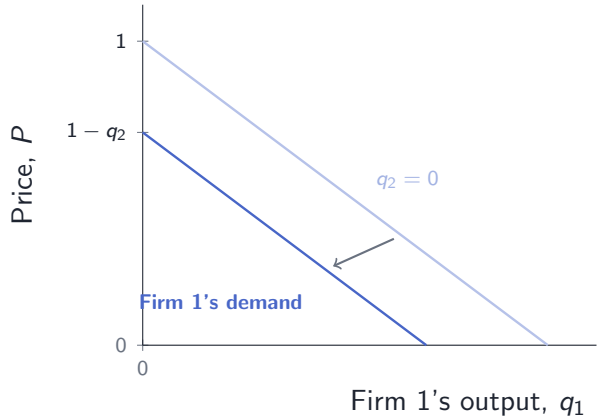
- Firm 1 takes firm 2's output q_2 as given and chooses q_1.

The demand left for firm 1

Rearrange the price equation:

$$\begin{aligned} P &= 1 - q_2 - q_1 \\ &= (1 - q_2) - q_1 \end{aligned}$$

- For a given q_2, this is a straight line in q_1.
- It starts at $1 - q_2$ and has slope -1.
- When q_2 rises, the line shifts down.



Firm 1 maximizes profit: MR = MC

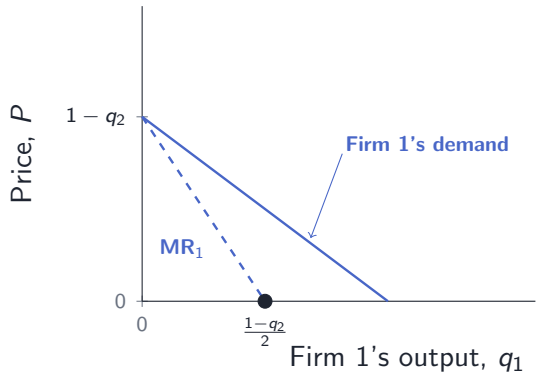
Firm 1's demand is a straight line starting at $1 - q_2$ with slope -1.

- By the Chapter 16 result, its MR starts at the same point and is twice as steep:

$$\text{MR}_1 = (1 - q_2) - 2q_1.$$

- Firm 1 sets $\text{MR}_1 = \text{MC} = 0$:

$$(1 - q_2) - 2q_1 = 0.$$



Firm 1's best-response function

Solve $(1 - q_2) - 2q_1 = 0$ for q_1:

$$2q_1 = 1 - q_2 \qquad \text{add } 2q_1 \text{ to both sides}$$

$$q_1 = \frac{1 - q_2}{2} \qquad \text{divide both sides by 2}$$

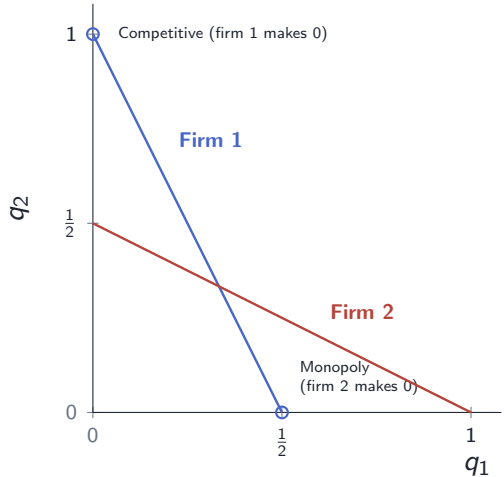
Definition: Best-response function

A firm's profit-maximizing output for each output its rival might choose.

- For each extra unit firm 2 produces, firm 1 produces half a unit less.
- **Firm 2**, by exactly the same steps: $q_2 = \frac{1 - q_1}{2}$.

The best responses on one graph

- Axes: firm 1's output across, firm 2's up.
- If firm 2 makes nothing, firm 1 makes the monopoly output, $\frac{1}{2}$.
- If firm 2 makes the competitive output, 1, firm 1 makes nothing.
- Firm 2's line is the mirror image.



The Nash equilibrium

- The mathematician and economist John Nash, winner of the 1994 Nobel Prize in Economics, defined an equilibrium that is now named after him.

Definition: Nash equilibrium

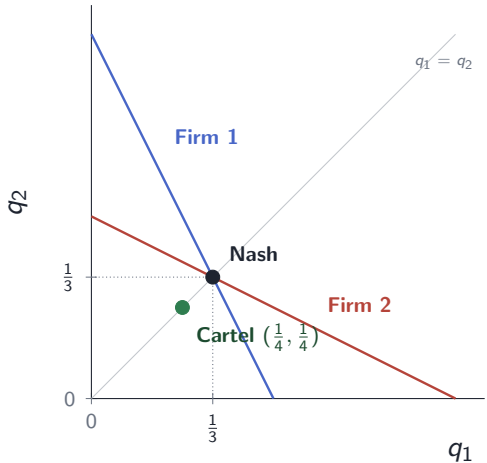
Each player in a game does the best it can, given that the other players are doing the best they can.

- In oligopoly, the players are the firms.
- Each firm does the best it can, given that the other firms are doing the best they can: each firm's output is its best response to the other's output.

Finding the Nash equilibrium

It is where the two lines cross. The firms are identical, so $q_1 = q_2 = q$:

$$q = \frac{1 - q}{2}$$
$$2q = 1 - q$$
$$q = \tfrac{1}{3}$$



Each firm produces more at the Nash equilibrium, $\frac{1}{3}$, than under the cartel, $\frac{1}{4}$.

Price and profit at the Nash equilibrium

$$Q = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \quad \text{total output}$$

$$P = 1 - \frac{2}{3} = \frac{1}{3} \quad \text{price from demand}$$

$$\pi = P \times q = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9} \quad \text{each firm's profit}$$

Finding

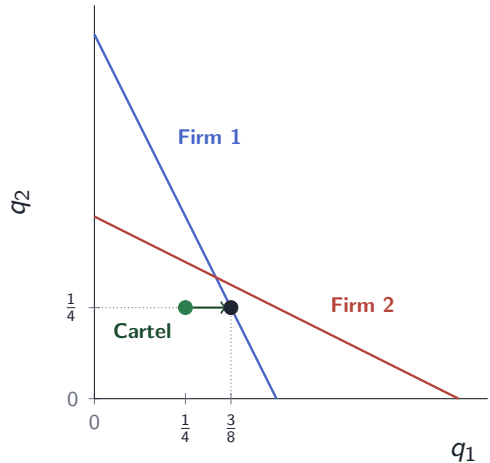
Each firm earns $\frac{1}{9}$, less than the $\frac{1}{8}$ it earns in the cartel.

Why the cartel breaks down

- Each firm asks: *given my rival's output, what output maximizes my profit?*
- The answer is its best response. Firm 2 produces its cartel share, $q_2 = \frac{1}{4}$, so

$$q_1 = \frac{1 - \frac{1}{4}}{2} = \frac{3}{8}.$$

- That is more than firm 1's cartel share, $\frac{1}{4} = \frac{2}{8}$.
- Firm 2 reasons the same way. Neither firm's agreed output is its best response.



What happens to profits

Firm 1 produces its best response, $\frac{3}{8}$. Firm 2 keeps to $\frac{1}{4} = \frac{2}{8}$.

$Q = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$	total output
$P = 1 - \frac{5}{8} = \frac{3}{8}$	price from demand
$\pi_1 = \frac{3}{8} \times \frac{3}{8} = \frac{9}{64}$	firm 1's profit
$\pi_2 = \frac{3}{8} \times \frac{2}{8} = \frac{6}{64}$	firm 2's profit

In the cartel each earned $\frac{1}{8} = \frac{8}{64}$.

Finding

Firm 1 gains ($\frac{9}{64} > \frac{8}{64}$). Firm 2 loses ($\frac{6}{64} < \frac{8}{64}$). Together they earn less ($\frac{15}{64} < \frac{16}{64}$).

The payoff table

Each firm chooses its cartel share, $\frac{1}{4}$, or its Nash output, $\frac{1}{3}$. Profits (firm 1, firm 2) are shown in 144ths, so $\frac{1}{8} = \frac{18}{144}$ appears as 18.

	Firm 2: cartel share	Firm 2: Nash output
Firm 1: cartel share	(18, 18)	(15, 20)
Firm 1: Nash output	(20, 15)	(16, 16)

- **How to read it:** the first number is firm 1's profit; the second is firm 2's.
- **For firm 1, compare within a column.** If firm 2 keeps its share: 20 beats 18. If firm 2 does not: 16 beats 15.
- So the larger output is better for firm 1 whatever firm 2 does. The same holds for firm 2.

Definition: Dominant strategy

A choice that is best whatever the other player does. Here: the larger output, for both firms.

The prisoners' dilemma

- Each firm's dominant strategy is the larger output.
- So both produce $\frac{1}{3}$: the Nash equilibrium. Each earns 16.
- Had both kept the agreement, each would earn 18.
- Each firm's own best choice leaves both worse off.

Definition: Prisoners' dilemma

A game in which each player's own best choice leaves both worse off than if they cooperated.

Finding

The cartel agreement serves the pair, but it is not each firm's best response. That is why collusion breaks down.

Part 3 summary

- Without a cartel, each firm maximizes its own profit, taking its rival's output as given.
- Best response: $q_1 = \frac{1-q_2}{2}$, and the same for firm 2.
- Nash equilibrium: each firm produces $\frac{1}{3}$ and earns $\frac{1}{9}$.
- Against a partner who keeps the cartel share, a firm's best response is $\frac{3}{8}$, more than $\frac{1}{4}$.
- Dominant strategy and prisoners' dilemma: both produce more, and both earn less.

Check: Part 3

Cartels are difficult to maintain because

A the monopoly output is very difficult to determine.

B the number of firms is always large.

C costs to the firms in a cartel are continually rising.

D each firm has an incentive to deviate from its agreed output level.

Check: Part 3 – answer

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Finding

Against a partner who keeps the cartel share, each firm's best response is a larger output.

Part 4

Market Outcomes Compared

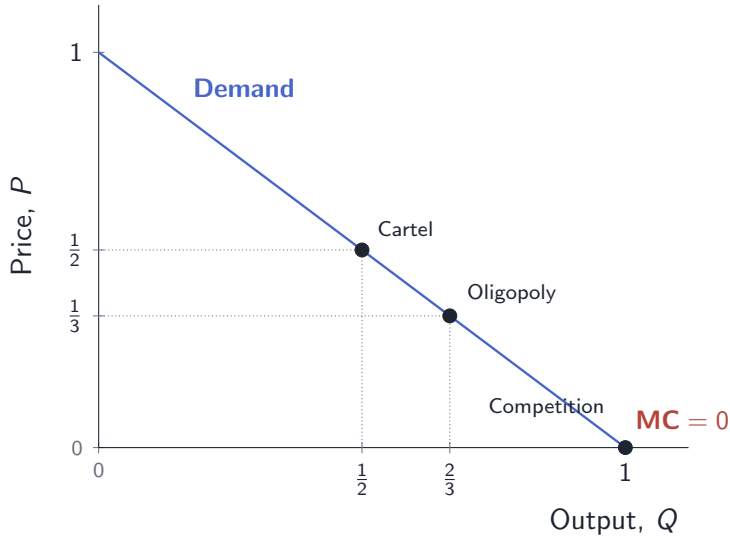
Three outcomes from the same demand and cost

	Total output	Price	Profit per firm	P and MC
Cartel (monopoly)	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{8}$	$P > \text{MC}$
Oligopoly (Nash)	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{9}$	$P > \text{MC}$
Perfect competition	1	0	0	$P = \text{MC}$

Finding

Output: $\frac{1}{2} < \frac{2}{3} < 1$. Price: $\frac{1}{2} > \frac{1}{3} > 0$. Oligopoly lies between monopoly and perfect competition.

The three outcomes on the graph



The general case

With demand $P = a - bQ$ and MC $= c$, the same steps give:

	Total output	Price	Profit per firm
Cartel (monopoly)	$\frac{a-c}{2b}$	$\frac{a+c}{2}$	$\frac{(a-c)^2}{8b}$
Oligopoly (Nash)	$\frac{2(a-c)}{3b}$	$\frac{a+2c}{3}$	$\frac{(a-c)^2}{9b}$
Perfect competition	$\frac{a-c}{b}$	c	0

Finding

Put $a = 1$, $b = 1$, $c = 0$ and each entry becomes the one in the simple model. The ordering is the same.

All four market structures in the long run

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Price and MC	$P = \text{MC}$	$P > \text{MC}$	$P > \text{MC}$	$P > \text{MC}$
Long-run profit	Zero	Zero	Can be positive	Can be positive
Why	Free entry	Free entry	Entry is difficult	Entry is blocked

Finding

Only perfect competition sets $P = \text{MC}$. In the other three, some buyers who value a unit above its cost do not get it.

Collusion and the law

- Collusion raises price and cuts output, so buyers lose.
- **Sherman Antitrust Act (1890):** agreements among competitors to fix prices or restrict output are illegal.
- **Clayton Act (1914):** individuals harmed by illegal agreements can sue to recover damages.
- **Example:** in 1996 Archer Daniels Midland agreed to plead guilty to fixing the prices of lysine and citric acid, two food and feed additives, and paid a $100 million fine.

Source: U.S. Department of Justice, press release, 15 October 1996.

Part 4 summary

- Cartel: output $\frac{1}{2}$, the lowest; price $\frac{1}{2}$, the highest.
- Oligopoly without collusion: output $\frac{2}{3}$; price $\frac{1}{3}$.
- Perfect competition: output 1; price 0, equal to MC.
- The general case gives the same ordering: oligopoly lies between monopoly and perfect competition.
- Price-fixing is illegal under the Sherman Act.

Check: Part 4

The equilibrium quantity in markets characterized by oligopoly is

A higher than in monopoly markets and higher than in perfectly competitive markets.

B higher than in monopoly markets and lower than in perfectly competitive markets.

C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

Check: Part 4 – answer

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Finding

In the model: $\frac{1}{2} < \frac{2}{3} < 1$.

Which rule answers which question

Question	Rule
What defines an oligopoly?	A few sellers, similar products, difficult entry, interdependence.
What does a cartel produce?	The monopoly output: MR = MC; $Q_K = \frac{1}{2}$ in the model.
What is a best response?	The profit-maximizing output, given the rival's output.
What is a Nash equilibrium?	Each firm's output is its best response to the other's.
Why does a cartel break down?	Each firm's best response to the cartel share is a larger output.
Where does oligopoly output lie?	Between monopoly and perfect competition.
Is collusion legal?	No: price-fixing is illegal under the Sherman Act.